Pre-course setup: your Python environment

Oct 24, 2026

Before Day 1, every participant opens the prep notebook, runs it to the end and reaches the line "You are ready for Day 1". It takes about 60 minutes, and nothing needs to be downloaded or installed. After "Before you start", this guide follows the notebook's five parts, with the same names and code, so you can keep both open side by side.

What you need

What	Details
A personal computer	Laptop or home computer, Windows or Mac. Not a hospital computer: you cannot sign in to a Google account there.
Google Chrome	Colab works in other browsers too, but this guide and its screenshots use Chrome.
A personal Google account	A private Gmail address. If you do not have one, create one for free at accounts.google.com. Do not use a work account.
An internet connection	The code runs on Google's servers, not on your computer.

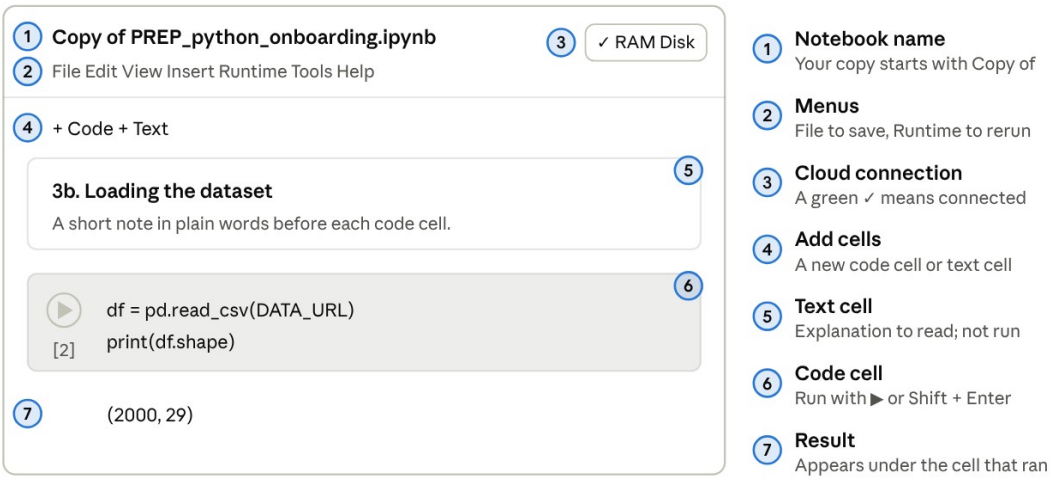
You do not need to install any software. Everything the course uses is already in Colab.

The course uses synthetic data only. Never upload real patient data to Colab, not even for practice.

What Colab is

Google Colab is a website that runs Python code in your browser. Every lesson in the course comes with a notebook: one page that mixes text cells with explanations and code cells that you run. The result of each code cell appears right under it.

A Colab notebook: text cells, code cells, and the result under each cell



The Colab screen · 7 parts

In every notebook you work from top to bottom: read a text cell, run the code cell after it, and check the result.

Before you start: open and save the notebook

1. Go to the course site, open the lesson "Pre-course preparation" and click **Open in Colab** next to the prep notebook. The notebook opens in a new tab.
2. If you are not signed in, click **Sign in** at the top right and sign in with your personal Google account.
3. Save your own copy: in the **File** menu choose **Save a copy in Drive**. A new tab opens with a notebook called "Copy of PREP_python_onboarding". Work only in this copy and close the original.

The copy is saved in your Google Drive, in the folder **Colab Notebooks**, where you will find it next time. Changes to the original notebook are not saved. Open every course notebook this way.

Part 1 · Running code

To run a code cell, click it and press **Shift + Enter**, or click the run button (▶️) at the left of the cell. The result appears under the cell, and the cursor moves to the next one.

- **The first run takes a few seconds.** Colab is assigning you a computer in the cloud. The top right shows "Connecting", then a green ✓ with RAM and Disk.
- **"Warning: This notebook was not authored by Google".** This appears the first time you run a notebook you did not write. The course notebooks are safe: click **Run anyway**.
- **Run cells top to bottom, in order.** Each cell uses what the cells above it created. The number in brackets at the left of a cell, for example [3], shows the order in which cells were run.

In the notebook, Part 1 uses Python as a calculator and stores values in variables to calculate a BMI.

Part 2 · Errors are part of the work

An error message is normal. Read its last line, fix the code and run the cell again. Nothing breaks. The notebook shows you a `NameError` on purpose, the most common error in the course; Common problems below lists the others.

Part 3 · Packages and the dataset

3a. Loading packages

A package is a set of ready-made tools that someone else wrote. The course uses five packages, and all of them are already installed in Colab:

Package	Used for
pandas	Tables: loading a file, filtering, counting
numpy	Calculations
matplotlib	Plots
statsmodels	Statistical models (regression)
scikit-learn	Machine-learning models

On a regular computer you would have to install them. In Colab you only load them into the notebook with `import`, once at the start of each notebook:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

The name after `as` is a standard short name: from now on you write `pd` instead of `pandas`. If the cell runs without a message, the packages are loaded. The prep notebook loads the first three; statsmodels and scikit-learn come in later lessons, and the readiness check in Part 5 confirms they are there. In every course notebook packages are loaded in the first code cell (Setup), so always start there.

3b. Loading the dataset

The course dataset is a CSV file of 2,000 synthetic admissions to internal-medicine wards. You do not need to download it: the notebook reads it straight from the internet.

```
DATA_URL = "https://raw.githubusercontent.com/Rotem-Gershon/cadets-
notebooks/main/data/cadets_admissions_release1_core.csv"

df = pd.read_csv(DATA_URL)
print(df.shape)
```

If everything worked, the result is `(2000, 29)`: 2,000 rows (one admission per row) and 29 columns. Now try it yourself (Your turn 3 in the notebook): add `df.head()` as a new line at the bottom of the same code cell and run the cell again. The first five rows of the dataset appear as a table.

Part 4 · Asking the dataset questions

Part 4 of the notebook asks the dataset a few clinical questions: a summary of one column, counts per category, an outcome rate, a filter to some of the patients, missing values and a first plot. Everything there builds on Parts 1 to 3.

Part 5 · Readiness check

The last cell of the prep notebook checks that the packages loaded and the dataset is complete. If it prints **You are ready for Day 1**, your setup is ready. If not, it prints what is missing, and the table below has the fix.

Common problems

What you see	What happened	What to do
NameError: name 'df' is not defined	A cell ran before the cells it depends on, or the connection dropped and the memory was cleared	Runtime → Run before runs every cell above the current one
Banner: Runtime disconnected	After about 90 minutes without activity Colab disconnects the cloud computer. Your code is kept	Reconnect, then Runtime → Run all
SyntaxError or invalid character	A missing quotation mark or bracket, or a character typed with a non-English keyboard layout	Switch the keyboard to English and compare with the original line, character by character
Your changes are gone after closing the notebook	You worked in the original, not in a copy	File → Save a copy in Drive before you start
HTTPError or URLError when loading the dataset	No internet connection, or the address changed	Check the connection and run the Setup cell again
The cell ran and nothing happened	A line that starts with # is a comment and does not run. Only the last line of a cell is shown automatically	Wrap what you want to see in print()
You cannot sign in to Google	A hospital computer or a work account	A personal computer and a private Gmail account

When nothing helps: Runtime → Restart session, then Runtime → Run all. This starts the notebook from scratch without deleting any code.

For a problem not listed here, send a screenshot of the message to [course support address].

Keyboard shortcuts

On a Mac, use Cmd instead of Ctrl.

Shortcut	What it does
Shift + Enter	Run the cell and move to the next one
Ctrl + Enter	Run the cell and stay in it
Ctrl + F9	Run the whole notebook (Run all)
Ctrl + M, then B	Add a new code cell below the current one
Ctrl + /	Turn a line into a comment (#) and back
Ctrl + Z	Undo the last change inside the cell